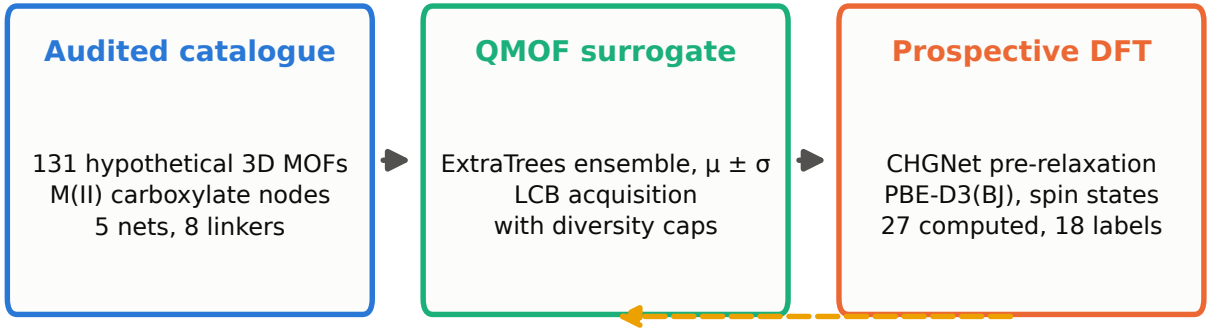
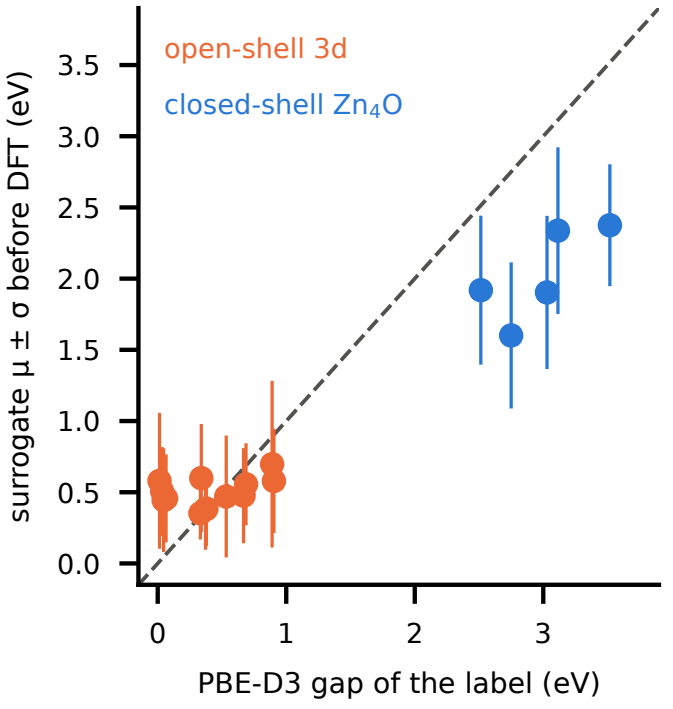



*labels retrain the surrogate · band-edge character · periodic GFN1-xTB check*

Open-shell Fe, Co, Cu: 13 of 13 below 1 eV; surrogate error 0.23 eV; metal d band edges

Closed-shell $Zn_4O$: surrogate 0.96 eV too low; linker band edges; periodic xTB within 0.23 eV

At this DFT budget: 3–4 × more low-gap hits than random selection, model error unchanged